

Notes: Levinsohn and Petrin (RES, 2003)

Estimating Production Functions Using Inputs to Control for Unobservables

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1 Big picture

- **Question.** How can we estimate production functions when firms choose inputs after observing their own productivity shocks, and when the investment proxy in Olley-Pakes is either unavailable or weak because many firms report zero investment?
- **Setting.** The paper studies Chilean manufacturing plants and asks whether *intermediate inputs* such as materials or electricity can proxy for the unobserved productivity state in the same spirit as investment in Olley and Pakes (1996).
- **Core challenge.** If firms with high productivity use more variable inputs, then input choices are correlated with the unobserved productivity term. Standard OLS and many traditional panel estimators therefore give biased production-function coefficients.
- **Main theoretical result.** If the intermediate-input demand function is strictly monotone in the transmitted productivity shock, conditional on capital, then the intermediate input can be inverted to recover the latent productivity state. This delivers a proxy-variable estimator that extends the Olley-Pakes logic.
- **Main implementation result.** The intermediate-input approach can be easier to use than the investment proxy because materials or electricity are almost always positive in the data, while investment is often zero. It may also work better when adjustment costs make investment respond only sluggishly to productivity shocks.
- **Main empirical result.** In Chilean data, the LP estimator usually gives lower coefficients on variable inputs and higher coefficients on capital than OLS, fixed effects, and lagged-input IV. This is exactly the pattern predicted by simultaneity bias.
- **Why this paper matters.** This paper is the classic justification for using intermediate inputs as proxies for unobserved productivity. It became a benchmark alternative to Olley-Pakes in empirical IO, trade, and productivity work because many datasets have persistent positive materials use but sparse investment.

2 Data and measurement (key objects)

2.1 Observed plant-level panel

- **Data source.** The paper uses the Chilean manufacturing census collected by the Instituto Nacional de Estadística (INE).
- **Sample.** The panel covers 6,665 plants from 1979 to 1986, including all plants with at least 10 employees.
- **Main industries.** To keep the empirical analysis manageable, the paper focuses on the four largest 3-digit industries: Food Products (ISIC 311), Metals (381), Textiles (321), and Wood Products (331).
- **Observed variables.** The dataset contains gross revenue (used as the output measure), capital, labor, and three intermediate inputs: materials, electricity, and fuels.
- **Labor detail.** Labor is split into blue-collar and white-collar workers, which the empirical specification interprets as unskilled and skilled labor.

- **Capital construction.** Capital is built from buildings, machinery, and vehicles using perpetual inventory with depreciation rates of 5%, 10%, and 20%, respectively.
- **Prices and deflation.** Gross revenue, capital, materials, electricity, and fuels are deflated to real 1980 Chilean pesos using annual deflators, mostly from the Banco Central de Chile.

2.2 Unobservables and timing

- **Productivity decomposition.** The key unobservable is a Hicks-neutral productivity shock, written as a transmitted component plus an i.i.d. component:

$$\epsilon_t = \omega_t + \eta_t.$$

- **Economic meaning.** The transmitted component ω_t is observed by the firm when it chooses inputs and therefore enters the firm’s decision rules. The innovation η_t is realized after current choices and is orthogonal to them.
- **State variables.** Capital is the main dynamic state variable. In the Olley-Pakes setup, investment is chosen with the goal of affecting future capital. In the LP setup, the intermediate input is not typically a state variable and can adjust much more freely.
- **Markov productivity.** The transmitted shock follows a first-order Markov process:

$$\omega_t = \mathbb{E}[\omega_t \mid \omega_{t-1}] + \xi_t,$$

where ξ_t is the innovation in productivity.

- **Timing implication.** Since the firm observes ω_t before choosing variable inputs, current labor and intermediate inputs are correlated with the latent productivity state. This is the source of simultaneity bias.

2.3 Proxy candidates and empirical applications

- **Proxy candidates.** The paper considers investment, materials, electricity, and fuels as potential proxies for productivity.
- **Why proxy choice matters.** A useful proxy should be positive for most plants in most years, should satisfy monotonicity with respect to productivity, and should be measured with limited error.
- **Empirical applications.** The paper uses the Chilean data to (i) estimate base-case production functions, (ii) compare materials and electricity as proxies, and (iii) compare LP with OLS, fixed effects, lagged-input IV, and Olley-Pakes.

3 Models and empirical specifications (all equations + interpretation)

3.1 The simultaneity problem and the sign of OLS bias

The general production environment is

$$Y_{it} = f(X_{it}, \epsilon_{it}; \beta), \tag{1}$$

where firms choose inputs X_{it} after observing productivity ϵ_{it} . In the simple two-input case,

$$y_{it} = \beta_0 + \beta_l l_{it} + \beta_k k_{it} + \epsilon_{it}. \quad (2)$$

The OLS coefficients can be written as

$$\hat{\beta}_l = \beta_l + \frac{\hat{\sigma}_{kk}\hat{\sigma}_{l\epsilon} - \hat{\sigma}_{lk}\hat{\sigma}_{k\epsilon}}{\hat{\sigma}_{ll}\hat{\sigma}_{kk} - \hat{\sigma}_{lk}^2}, \quad (3)$$

$$\hat{\beta}_k = \beta_k + \frac{\hat{\sigma}_{ll}\hat{\sigma}_{k\epsilon} - \hat{\sigma}_{lk}\hat{\sigma}_{l\epsilon}}{\hat{\sigma}_{ll}\hat{\sigma}_{kk} - \hat{\sigma}_{lk}^2}. \quad (4)$$

- **Interpretation.** If labor responds more strongly than capital to productivity shocks, then OLS tends to overstate the labor coefficient and understate the capital coefficient.
- **Why this matters for the paper.** This is exactly the empirical pattern the paper later finds when comparing OLS to the intermediate-input proxy estimator.

3.2 The Olley-Pakes investment proxy

Olley and Pakes begin from the log production function

$$y_t = \beta_0 + \beta_l l_t + \beta_k k_t + \omega_t + \eta_t. \quad (5)$$

If investment is a monotone function of the state variables,

$$i_t = i_t(\omega_t, k_t), \quad (6)$$

then it can be inverted to recover productivity,

$$\omega_t = \omega_t(i_t, k_t). \quad (7)$$

Substituting this into production yields the first-stage equation

$$y_t = \beta_l l_t + \phi_t(i_t, k_t) + \eta_t, \quad (8)$$

where

$$\phi_t(i_t, k_t) = \beta_0 + \beta_k k_t + \omega_t(i_t, k_t). \quad (9)$$

The second stage uses the Markov property of productivity and the fact that capital does not respond to the current innovation:

$$y_t^* = y_t - \beta_l l_t = \beta_0 + \beta_k k_t + \mathbb{E}[\omega_t \mid \omega_{t-1}] + \eta_t^*, \quad (10)$$

where

$$\eta_t^* = \xi_t + \eta_t. \quad (11)$$

- **Interpretation.** Investment proxies the latent state ω_t , which removes the correlation between labor and productivity in the first stage.
- **Practical problem.** This strategy is only available for plants with positive investment. When many plants report zero investment, the effective sample collapses.

3.3 Intermediate inputs as proxies

Levinsohn and Petrin add a second flexible input, denoted here by ι_t , which can be materials or energy. The log production function becomes

$$y_t = \beta_0 + \beta_l l_t + \beta_k k_t + \beta_\iota \iota_t + \omega_t + \eta_t. \quad (12)$$

The intermediate-input demand function is

$$\iota_t = \iota_t(\omega_t, k_t). \quad (13)$$

If it is monotone in ω_t , it can be inverted as

$$\omega_t = \omega_t(\iota_t, k_t). \quad (14)$$

The first-stage equation is then

$$y_t = \beta_l l_t + \phi_t(\iota_t, k_t) + \eta_t, \quad (15)$$

with

$$\phi_t(\iota_t, k_t) = \beta_0 + \beta_k k_t + \beta_\iota \iota_t + \omega_t(\iota_t, k_t). \quad (16)$$

The second stage becomes

$$y_t^* = y_t - \beta_l l_t = \beta_0 + \beta_k k_t + \beta_\iota \iota_t + \mathbb{E}[\omega_t \mid \omega_{t-1}] + \eta_t^*. \quad (17)$$

- **Key difference from OP.** The proxy itself, ι_t , is now also a direct production input. So unlike investment, its coefficient must be identified in the second stage.
- **Identification logic.** Since the current intermediate input responds to the productivity innovation, it is generally correlated with η_t^* . But lagged intermediate input ι_{t-1} is chosen before the current innovation and can therefore serve as a moment condition.

3.4 The monotonicity condition

For the intermediate input to be a valid proxy, more productive firms must use more of it conditional on capital. The paper shows that

$$\text{sign}\left(\frac{\partial \iota}{\partial \omega}\right) = \text{sign}(f_{\iota l} f_{l \omega} - f_{l \iota} f_{\omega}). \quad (18)$$

- **Interpretation.** The sign of the response of intermediate-input demand to productivity depends on how productivity changes the marginal product of labor and the intermediate input.
- **When it holds.** The condition is easy to verify for common technologies such as Cobb-Douglas and CES. This is one theoretical advantage relative to the investment proxy, where monotonicity comes from a more involved dynamic argument.
- **Economic intuition.** If higher productivity raises the marginal product of the intermediate input, firms expand output and use more materials or energy.

3.5 Empirical production-function specification

In the empirical implementation, the gross-output production function is approximated by a Cobb-Douglas technology:

$$y_t = \beta_0 + \beta_s z_t^s + \beta_u z_t^u + \beta_e e_t + \beta_f f_t + \beta_m m_t + \beta_k k_t + \omega_t + \eta_t, \quad (19)$$

where z_t^s and z_t^u denote skilled and unskilled labor, while e_t , f_t , and m_t are electricity, fuels, and materials.

Using materials as the proxy, the first stage is

$$y_t = \beta_s z_t^s + \beta_u z_t^u + \beta_e e_t + \beta_f f_t + \phi_t(m_t, k_t) + \eta_t, \quad (20)$$

with

$$\phi_t(m_t, k_t) = \beta_0 + \beta_m m_t + \beta_k k_t + \omega_t(m_t, k_t). \quad (21)$$

Rather than use a polynomial approximation exactly as in OP, the paper estimates conditional moments such as $\mathbb{E}[y_t | k_t, m_t]$ by local quadratic regression, and then subtracts them out to obtain

$$y_t - \mathbb{E}(y_t | k_t, m_t) = \beta_s(z_t^s - \mathbb{E}(z_t^s | k_t, m_t)) + \beta_u(z_t^u - \mathbb{E}(z_t^u | k_t, m_t)) \quad (22)$$

$$+ \beta_e(e_t - \mathbb{E}(e_t | k_t, m_t)) + \beta_f(f_t - \mathbb{E}(f_t | k_t, m_t)) + \eta_t. \quad (23)$$

- **What this stage identifies.** The first stage recovers the coefficients on the flexible non-proxy inputs: skilled labor, unskilled labor, electricity, and fuels.
- **Why capital and materials remain for stage 2.** Capital enters both production and the proxy function, and materials are both a proxy and a direct input. So these coefficients are not yet separately identified.

3.6 Second-stage moments, GMM, and bootstrap inference

Define $\beta^* = (\beta_m, \beta_k)$. The second-stage residual is written as

$$\eta_t^*(\beta^*) = y_t - \beta_s z_t^s - \beta_u z_t^u - \beta_e e_t - \beta_f f_t - \beta_m m_t - \beta_k k_t - \mathbb{E}[\omega_t | \omega_{t-1}]. \quad (24)$$

The paper uses the moment conditions

$$\mathbb{E}[(\xi_t + \eta_t)k_t] = 0, \quad \mathbb{E}[(\xi_t + \eta_t)m_{t-1}] = 0, \quad (25)$$

and, more generally, an instrument vector

$$Z_t = \{k_t, m_{t-1}, z_{t-1}^s, z_{t-1}^u, e_{t-1}, f_{t-1}, k_{t-1}, m_{t-2}\}. \quad (26)$$

The coefficients are estimated by minimizing a GMM criterion of the form

$$Q(\beta^*) = \min_{\beta^*} \sum_i \sum_h \sum_t [\hat{\eta}_{it}^*(\beta^*) Z_{iht}]^2. \quad (27)$$

- **Interpretation.** Capital is predetermined, so it is orthogonal to the current productivity innovation. Lagged materials identify the coefficient on the proxy input because they are correlated with current materials but uncorrelated with the current innovation.
- **Inference.** The estimator has many generated regressors and repeated nonparametric steps, so the paper uses a firm-level bootstrap to conduct inference and to compare estimators.
- **Alternative estimator.** The paper also implements the Blundell-Bond system GMM estimator, but finds it extremely unstable in these data, with estimates often landing on the boundary of the admissible parameter space.

4 Identification and instruments (what makes the estimates credible)

4.1 Why the investment proxy may fail

- **Zero-investment problem.** The Olley-Pakes investment proxy requires positive investment in order to invert the investment function. In the Chilean data, only 35.9% to 44.8% of observations have nonzero investment depending on the industry.
- **Efficiency loss.** This means that more than half the sample would be discarded if one relied on investment alone.
- **Adjustment-cost problem.** If investment has nonconvex adjustment costs, the investment demand function may have kinks. Then investment may not fully respond to productivity shocks, and correlation between regressors and the error term can remain.

4.2 What identifies the LP model

- **Monotone input demand.** The intermediate-input demand function must be strictly monotone in productivity conditional on capital, so the proxy can be inverted.
- **Flexible timing.** The intermediate input is typically not a state variable, so it can respond more smoothly to productivity than investment does.
- **Predetermined capital.** Current capital does not respond to the current innovation in productivity, which gives the key moment for identifying β_k .
- **Lagged proxy moment.** Lagged materials are correlated with current materials because of persistence but are not contaminated by the current innovation in productivity. This identifies β_m .
- **Common input-price assumption.** The demand function is written as $\iota_t = \iota_t(\omega_t, k_t)$, which suppresses firm-specific input prices. If prices vary systematically across time, region, or firm type, the demand function must be adjusted accordingly.

4.3 How the paper chooses among proxies

The paper proposes three practical specification tests for choosing the proxy.

- **Test 1: Monotonicity plots.** Plot the implied productivity proxy as a function of capital and the candidate intermediate input. A valid proxy should yield a function that is increasing in the proxy input conditional on capital.
- **Test 2: Proxy invariance.** Estimate the model using different candidate proxies, especially materials and electricity. If the model is correctly specified and both proxies are valid, the resulting coefficients on the non-proxy inputs should be similar.
- **Test 3: Over-identifying restrictions.** Inputs lagged one period should be uncorrelated with next period's innovation in productivity. The paper extends the Olley-Pakes logic by testing this with six over-identifying conditions.

TABLE 1
Some descriptive statistics on Chilean manufacturing

Year	GDP	Industry (ISIC)							
		Food products (311)		Metals (381)		Textiles (321)		Wood products (331)	
		Plants	Value added	Plants	Value added	Plants	Value added	Plants	Value added
1979	997.6	1537	39.0	459	10.0	503	12.4	524	10.4
1980	1075.3	1439	43.4	447	11.0	445	12.9	449	8.7
1981	1134.7	1351	42.7	413	11.5	403	11.3	406	6.8
1982	974.9	1319	47.0	365	8.1	350	8.7	358	6.5
1983	968.0	1297	42.9	322	8.3	327	9.7	335	8.1
1984	1029.4	1340	46.8	358	11.4	336	10.4	339	10.3
1985	1054.6	1338	49.1	351	9.6	337	10.8	342	10.1
1986	1114.3	1288	61.4	347	9.6	331	12.9	313	5.3

Note: GDP figures from the *International Financial Statistics Yearbook*. GDP and value added in millions of 1980 pesos.

TABLE 2
Per cent of non-zero observations

Industry (ISIC)	Investment	Fuels	Materials	Electricity
Food products (311)	42.7	78.0	99.8	88.3
Metals (381)	44.8	63.1	99.9	96.5
Textiles (321)	41.2	51.2	99.9	97.0
Wood products (331)	35.9	59.3	99.7	93.8

4.4 Relation to the literature

- **Relative to Olley-Pakes.** LP keeps the proxy-variable logic but swaps investment for intermediate inputs. The gain is mostly practical: less truncation and often a more responsive proxy.
- **Relative to OLS, fixed effects, and lagged-input IV.** Those estimators do not solve the same simultaneity problem under the same behavioral assumptions, and the paper's empirical comparisons show they can differ sharply from LP.
- **Relative to Blundell-Bond.** The paper treats system GMM as a useful alternative in principle, but in practice the estimator is too unstable in this application to be informative.

5 Tables and figures

5.1 Table 1: Descriptive statistics on Chilean manufacturing

Read:

- The table shows a sharp macroeconomic cycle around the 1982–1983 debt crisis: Chilean GDP falls from 1,134.7 in 1981 to 974.9 in 1982 and remains weak in 1983 before recovering.
- Plant counts fall in every industry over the sample. For example, Food Products declines from 1,537 plants in 1979 to 1,288 in 1986, and Wood Products from 524 to 313.
- These patterns highlight that the sample is unbalanced and contains meaningful exit and consolidation, which is important for any production-function estimator.
- The value-added series also shows that the industries experience common macroeconomic shifts, which motivates allowing the proxy function to vary across subperiods.

5.2 Table 2: Percentage of nonzero observations

Read:

- Investment is positive for only 35.9% to 44.8% of plant-year observations, depending on the industry.
- Materials are positive for 99.7% to 99.9% of observations, and electricity is positive for 88.3% to 97.0%.
- Fuels are used more often than investment but still much less often than materials, with nonzero shares ranging from 51.2% to 78.0%.
- This table is the main empirical reason for preferring intermediate-input proxies to investment in these data: OP would lose more than half the sample immediately.

5.3 Table 3: Base-case parameter estimates for four industries

Read:

- Using materials as the proxy, the estimated material coefficient is the largest input elasticity in every industry: 0.500 in Food Products, 0.420 in Metals, 0.500 in Textiles, and 0.550 in Wood Products.
- Capital is typically the next most important factor, with coefficients between 0.18 and 0.29.
- The skilled- and unskilled-labor coefficients are positive and statistically significant in almost all cases. Electricity is small, and in Textiles it is the only input that is not precisely estimated.
- Estimated returns to scale range from 1.007 to 1.172, so the technologies are close to constant returns even though some point estimates are slightly above one.

TABLE 3
Base case parameter estimates for four industries
(bootstrapped standard errors in parentheses)

Input	Industry (ISIC code)			
	311	381	321	331
Unskilled labour	0.139 (0.010)	0.172 (0.033)	0.130 (0.024)	0.193 (0.034)
Skilled labour	0.051 (0.009)	0.188 (0.025)	0.155 (0.026)	0.133 (0.030)
Electricity	0.085 (0.007)	0.081 (0.015)	0.005 (0.019)	0.047 (0.021)
Fuels	0.023 (0.004)	0.020 (0.011)	0.038 (0.010)	0.021 (0.014)
Materials	0.500 (0.078)	0.420 (0.091)	0.500 (0.118)	0.550 (0.086)
Capital	0.240 (0.053)	0.290 (0.094)	0.180 (0.095)	0.190 (0.090)
Returns to scale	1.037 (0.059)	1.172 (0.075)	1.007 (0.113)	1.133 (0.157)
No. obs.	6115	1394	1129	1032

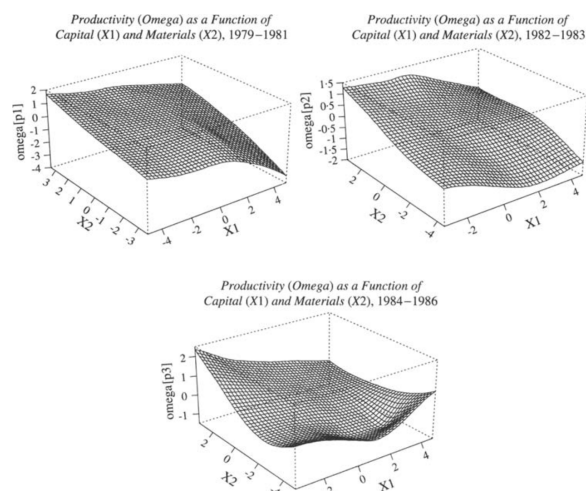


FIGURE 1
Productivity as a function of capital and materials use

TABLE 4
Two specification tests for the proxy choice electricity and materials

Difference in point estimates	Industry (ISIC code)			
	311	381	321	331
Unskilled labour	-0.002 (0.005)	-0.004 (0.009)	-0.001 (0.012)	0.010 (0.014)
Skilled labour	-0.003 (0.003)	-0.004 (0.008)	-0.011 (0.012)	0.006 (0.011)
Fuels	0.000 (0.002)	0.004 (0.004)	0.002 (0.004)	-0.005 (0.006)
Capital	0.03 (0.075)	-0.03 (0.131)	0.07 (0.145)	0.13 (0.109)
Number sig. different at 10%	0	0	0	0
<i>P</i> -value for over-identification test				
Materials	0.01	0.51	0.86	0.68
Electricity	0.00	0.32	0.10	0.40

Note: The first panel lists the difference in the coefficients on the non-proxy inputs using electricity vs. materials as the proxy (electricity estimates minus materials estimates). Bootstrap standard errors of the differences are reported in parentheses. The number of these differences that are statistically different from zero at a level of significance of 0.1 is computed using a symmetric 90% confidence interval implied by the bootstrap. The second panel lists the *P*-value for the test of the over-identifying restrictions for each proxy in each industry.

5.4 Figure 1: Productivity as a function of capital and materials use

Read:

- The figure plots the smoothed proxy function for Textiles (ISIC 321) across three macro periods: 1979–1981, 1982–1983, and 1984–1986.
- In the first two periods, productivity appears to increase monotonically in materials for essentially all capital levels, which is exactly what the theory requires.
- In the third period, there are some local violations of monotonicity at low materials use and high capital.
- The paper’s interpretation is that monotonicity broadly holds in the three largest industries, but in smaller industries or later subperiods one may need a richer specification or more disaggregated price controls.

5.5 Table 4: Specification tests for the proxy choice, electricity versus materials

Read:

- The top panel compares coefficient estimates when electricity rather than materials is used as the proxy. None of the coefficient differences is statistically significant at the 10% level.
- The basic message is that materials and electricity deliver very similar first-stage estimates, which supports the idea that both can proxy the same productivity state.
- The bottom panel reports over-identification tests. In three of the four industries, the over-identifying restrictions are not rejected at conventional levels.
- The exception is the largest industry, Food Products (311), where the *p*-values are 0.01 for materials and 0.00 for electricity. The paper suggests that, in this industry, a more flexible first-stage specification may be needed.

5.6 Figure 2 and Table 5: OLS versus LP

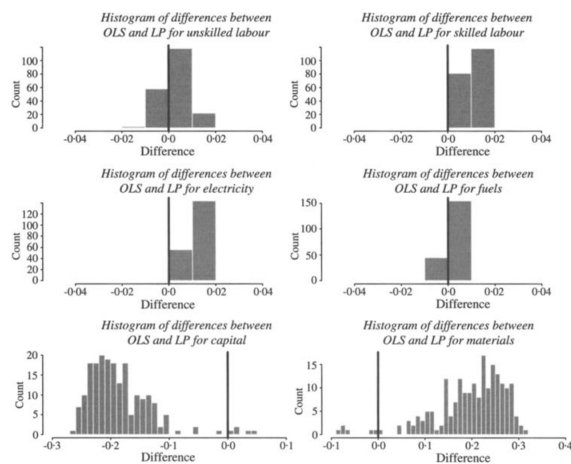


FIGURE 2
Differences between OLS and LP estimates

TABLE 5
OLS estimate minus the LP estimate

Input	Industry (ISIC code)			
	311	381	321	331
Unskilled labour				
Difference	0.004	0.001	-0.010	-0.009
% > 0	70	51	21	27
Skilled labour				
Difference	0.011	-0.008	-0.003	0.013
% > 0	100	23	48	84.5
Electricity				
Difference	0.011	0.005	0.005	0.007
% > 0	100	74.5	83.5	68.5
Fuels				
Difference	0.001	0.004	0.002	-0.004
% > 0	77.5	81	61	20.5
Materials				
Difference	0.210	0.166	0.197	0.075
% > 0	97	76.5	84.5	88.5
Capital				
Difference	-0.190	-0.225	-0.141	-0.112
% > 0	2.5	14.5	20	24
Returns to scale				
Difference	0.049	-0.056	0.049	-0.030
% > 0	78.0	25.0	64.5	56.5
No. obs.	6115	1394	1129	1032

Note: The top entry in each cell is the difference between the OLS and LP estimate. The bottom entry in each cell is the percentage of the 200 bootstrapped samples that yielded estimates in which the OLS coefficient was larger than the LP coefficient (i.e. $\hat{\beta}_{OLS} - \hat{\beta}_{LP} > 0$).

Read:

- For Food Products (311), Figure 2 shows that OLS usually estimates larger coefficients on variable inputs and a smaller coefficient on capital than LP.
- Table 5 shows this is not just a one-industry fact. For materials, the OLS minus LP difference is 0.210 in Food Products, 0.166 in Metals, 0.197 in Textiles, and 0.075 in Wood Products.
- Capital goes in the opposite direction in every industry: the OLS minus LP difference is -0.190, -0.225, -0.141, and -0.112, respectively.
- Overall, 19 of the 24 coefficient comparisons across industries are consistent with the simultaneity story. OLS overstates variable-input elasticities and understates capital.

5.7 Figure 3: Olley-Pakes versus Levinsohn-Petrin

Read:

- Figure 3 compares OP and LP in Food Products (311). The general pattern looks similar to the OLS-versus-LP comparison, although the unskilled-labor difference is less dramatic.
- The important message is that the intermediate-input proxy appears to respond more fully to the productivity shock than investment does.

- This is exactly the economic argument behind the paper: materials or electricity may give a cleaner signal of productivity in datasets where investment is lumpy.

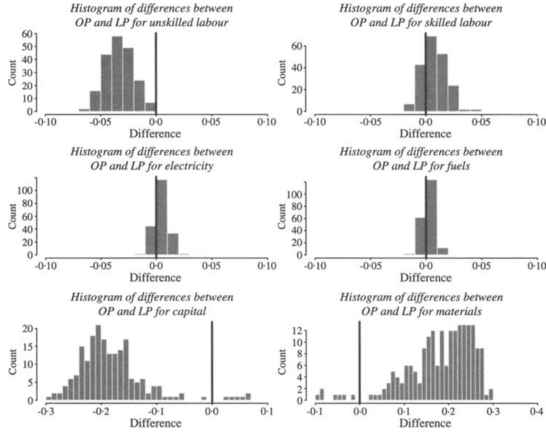


FIGURE 3
Differences between OP and LP estimates

TABLE 6
Comparisons across estimators P -value for $H_0: \beta_1 = \beta_2$

Comparison	Industry (ISIC code)			
	311	381	321	331
Levinsohn–Petrin vs.				
OLS	<0.01	0.20	0.58	0.21
Fixed effects	<0.01	<0.01	<0.01	<0.01
Instrumental variables	<0.01	0.22	0.09	<0.01
Olley–Pakes	<0.01	0.54	0.20	0.89
Levinsohn–Petrin ($i > 0$ only)	<0.01	0.02	0.27	0.93
Olley–Pakes vs.				
OLS	<0.01	0.04	0.19	0.46
Fixed effects	<0.01	<0.01	<0.01	<0.01
Instrumental variables	<0.01	<0.01	<0.01	<0.01
Levinsohn–Petrin ($i > 0$ only)	0.56	0.47	0.85	0.55
Fixed effects vs.				
OLS	<0.01	<0.01	<0.01	<0.01
Instrumental variables	<0.01	<0.01	<0.01	<0.01
No. obs.	6115	1394	1129	1032

Note: The cells in the table contain the P -value for a standard Wald test for “no differences between the (vector of) parameter estimates for estimators 1 and 2”. <0.01 indicates a P -value that is less than 0.01.

5.8 Table 6: Comparisons across estimators

Read:

- In the largest industry, Food Products (311), LP rejects equality with every other reported estimator at the 1% level: OLS, fixed effects, lagged-input IV, OP, and even LP re-estimated on the positive-investment subsample.
- Across industries, LP strongly disagrees with fixed effects in all four industries and with lagged-input IV in three of the four.
- OP and LP restricted to firms with positive investment are the most compatible pair, which makes sense because both rely on the same positive-investment subsample and similar persistence logic.
- The fixed-effects estimator is the biggest outlier in the table: it rejects compatibility with every other estimator in every industry at the 1% level. The paper interprets this as evidence that the persistent productivity shock varies within firms over time and cannot be treated as a simple fixed effect.

6 Short summary

- **Conclusion.** Intermediate inputs can proxy for unobserved productivity in production-function estimation when they respond monotonically to productivity conditional on capital.
- **Identification insight.** The coefficient on the proxy input is identified in the second stage by using lagged proxy values as instruments, while capital is identified from its predetermined timing.

- **Empirical lesson.** In Chilean manufacturing, LP yields coefficient estimates that differ sharply from OLS, fixed effects, and lagged-input IV in exactly the way one would expect if standard estimators were contaminated by simultaneity bias.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that intermediate inputs provide a practical and theoretically coherent alternative to investment as proxy variables for unobserved productivity.
- It shows that the logic of Olley-Pakes can be extended to a setting in which the proxy itself enters production, as long as one uses lagged proxy values to identify its coefficient.
- It also demonstrates that proxy choice is an empirical issue. Materials or electricity can work well, but they must be checked against monotonicity and over-identification diagnostics.

7.2 Economic intuition

- When a plant learns it has a high productivity shock, it wants to expand output. The most immediate way to do so is often to hire more labor and use more materials or energy.
- Investment may react too slowly because of adjustment costs and zeros, but intermediate inputs often adjust on the spot. That makes them attractive state proxies.
- The reason OLS fails is simple: high-productivity plants look like they have high marginal products of labor and materials, so naive regressions attribute too much output to variable inputs and too little to the latent productivity state.

7.3 Takeaway

This paper is best read as a practical extension of Olley-Pakes. Its main contribution is not a radically different theory of production, but a more usable proxy-variable strategy for the kinds of plant-level panels that economists actually have. When investment is frequently zero or only weakly responsive to shocks, materials and electricity can rescue the proxy approach. The empirical payoff is substantial: production functions look very different once simultaneity is handled correctly, and the differences line up with clear economic intuition about how firms respond to productivity shocks.